

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

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COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 50

Code of Conduct

I sanjay kumar soy(192311151) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *sanjay kumar soy*

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-	Covers most expected points	Covers only basic points	Very limited coverage

		points completely			
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Cloud computing is the delivery of computing services such as servers, storage, databases, networking, and software over the Internet. Instead of purchasing and maintaining physical hardware, users can access these resources on demand and pay only for what they use.

Characteristics of Cloud Computing

1. On-Demand Self-Service

Users can access computing resources whenever required without depending on the service provider. For example, a company can instantly create a virtual server on Amazon Web Services (AWS).

2. Broad Network Access

Cloud services are available over the Internet and can be accessed from anywhere using devices such as laptops, smartphones, or tablets. For example, students can access Google Drive from any location.

3. Resource Pooling

Cloud providers share a common pool of computing resources among multiple users while keeping each user's data secure. This improves resource utilization and reduces costs. For example, Microsoft Azure serves multiple organizations using shared infrastructure.

4. Rapid Elasticity (Scalability)

Cloud resources can be increased or decreased quickly based on user demand. For example, an e-commerce website can automatically add more servers during a festive sale and reduce them after the traffic decreases.

Difference from Traditional On-Premise Computing

Unlike traditional on-premise computing, where organizations must purchase, install, and maintain their own hardware and software, cloud computing provides scalable, flexible, and cost-effective resources over the Internet with minimal infrastructure management.

2. Trace the Evolution of Cloud Computing from Hardware Evolution to Internet Software Evolution, and Show How These Changes Enabled Modern Cloud Platforms.

Cloud computing evolved through several stages of technological advancement:

1. Hardware Evolution:

Initially, organizations used expensive mainframe computers. Later, powerful servers and virtualization technology allowed multiple virtual machines to run on a single physical server, improving resource utilization.

2. Network Evolution:

The growth of the Internet and high-speed communication networks enabled users to access computing resources remotely from anywhere in the world.

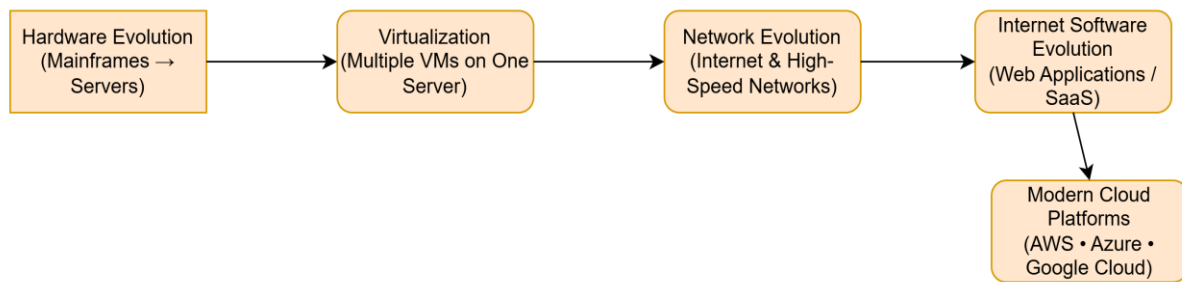
3. Internet Software Evolution:

Software changed from being installed on individual computers to being accessed through web

browsers. This led to Software as a Service (SaaS), such as Google Workspace and Microsoft 365, where applications are available online.

4. Modern Cloud Platforms:

The combination of virtualization, high-speed Internet, and web-based software enabled cloud platforms such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP). These platforms provide Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS), allowing users to access scalable and cost-effective computing resources on demand.



3. Explain the Role of Server Virtualization in Cloud Computing. Differentiate Between Virtualization as an Enabling Technology and Cloud Computing as a Service Model.

Server virtualization is the process of creating multiple virtual machines (VMs) on a single physical server using a hypervisor. It improves hardware utilization, reduces costs, and allows resources to be allocated efficiently.

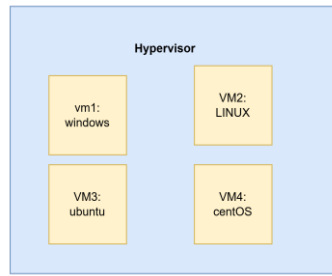
Role in Cloud Computing:

- Allows multiple virtual machines to run on one physical server.
- Increases resource utilization and reduces hardware costs.
- Supports easy scaling and flexible resource allocation.
- Forms the foundation of cloud platforms such as AWS and Microsoft Azure.

Difference:

- Virtualization is an enabling technology that creates and manages virtual machines on physical hardware.
- Cloud computing is a service model that delivers computing resources (IaaS, PaaS, SaaS) over the Internet using technologies like virtualization.

Example: VMware is a virtualization technology, while AWS is a cloud computing platform.



4. Compare SOAP-based and REST-based Web Services in Terms of Communication Style, Data Exchange, and Suitability for Cloud Applications.

Feature	SOAP	REST
Communication Style	Protocol-based with strict standards	Architectural style using HTTP methods (GET, POST, PUT, DELETE)
Data Exchange	Uses only XML	Supports JSON, XML, and other formats (JSON is most common)
Suitability for Cloud Applications	Best for secure enterprise applications like banking	Best for cloud and web applications because it is lightweight, fast, and scalable

5. Differentiate IaaS, PaaS, SaaS, and XaaS with One Suitable Real-World Example for Each Model.

IaaS (Infrastructure as a Service):

Provides virtualized computing resources such as servers, storage, and networking over the Internet. Users manage the operating system and applications.

Example: Amazon Web Services (AWS EC2).

PaaS (Platform as a Service):

Provides a platform for developers to build, test, and deploy applications without managing the underlying infrastructure.

Example: Google App Engine.

SaaS (Software as a Service):

Provides ready-to-use software applications over the Internet. Users only access the application without worrying about installation or maintenance.

Example: Microsoft 365.

XaaS (Anything as a Service):

A broad term that refers to delivering any IT service over the Internet, including infrastructure, platforms, software, storage, and databases.

Example: Google Firebase (Backend as a Service).